University of East London

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SCHOOL OF ARCHITECTURE, COMPUTING & ENGINEERING

Submission instructions

- Cover sheet to be attached to the front of the assignment when submitted
- All pages to be numbered sequentially
- All work must be uploaded in a ready to submit state.
- It must have Assignment cover sheets before submission.

Module code	**CN6001 and CN5023**		
Module title	Enterprise Architecture and Cloud Computing		
Module leader	Dr. Aloysius Edoh PhD		
Assignment tutor	Ragul, Tobi, Mayank Trivedi, Afrim, Mustak and Dr Chris O'K		
Assignment title	"Design and implementation of Web based Portal using Enterprise Cloud Architecture with CASE Tools"		
Assignment number	One		
Weighting	30%		
Handout date	**10th March 2026**		
Submission date	**1st /May/2026**		
Learning outcomes assessed by this assignment	1. Evaluate the relation between Enterprise Portal and mobile application. 2. Evaluate the issues in Enterprise systems including benefits and drawbacks. 3. Critically analyse the risk, security and protection in information system design and Enterprise Portal deployment *Subject-based practical skills* 4. Implement, simulate and test prototype cloud web-based Enterprise Portal using appropriate software packages and or platforms; *Skills for life and work (general skills)* 5. Group work using software packages, documentation/report writing skills and working to meet deadlines;		
Turnitin submission requirement	**Yes**	**Turnitin GradeMark feedback used?**	**Yes**
Moodle Grade Book submission used?	**Yes**	**Moodle Grade Book feedback used?**	**Yes**
Other electronic system used?	**Yes**	**Are submissions / feedback totally electronic?**	**No**
Additional information	**There are both electronic submission and feedbacks.**		

Form of assessment:

☐ Individual work ☒ Group work

For **group work** assessment which requires members to submit both individual and group work aspects for the assignment, the work should be submitted as:

☐ Consolidated single document ☐ Separately by each member

Number of assignment copies required:

☒ 1 ☐ 2 ☐ Other

Assignment to be presented in the following format:

☒ On-line submission
☐ Stapled once in the top left-hand corner
☐ Glue bound
☐ Spiral bound
☐ Placed in a A4 ring bound folder (not lever arch)

Note: To students submitting work on A3/A2 boards, work must be contained in suitable protective case to ensure any damage to work is avoided.

Soft copy:

☐ CD (to be attached to the work in an envelope or purpose made wallet adhered to the rear)
☐ USB (to be attached to the work in an envelope or purpose made wallet adhered to the rear)
☐ Soft copy when required

Coursework: Research with Practical Component on the "Design and implementation of Web Portal with Enterprise Cloud Architecture CASE Tools"

Assessment
Due Date: 1st /May/2026

This Group work is a Research work with a Practical Component on the "Design and Implementation of Web Portal using Enterprise Cloud Architecture CASE Tools".

1.0 Course Work and Assessment:

This group work constitutes of 30% of the overall module assessment. The assignment must be carried out by group basis with a max of 3 to 4 students in a group. Note the following:

Part A
The first part of the group work is an investigation, design and modelling of an Enterprise Cloud Architecture System for a selected case study using Bonita software, StarUML and or Yworks CASE TOOLS and the documentation of the work carried out.

Part B
The second part of the course work is the implementation of the prototype enterprise **portal** software for the selected case study using any suitable Web/ Cloud platform for the web interface linked to database backend. This section must include an evaluation of the cloud portal technology and techniques used for the design and implementation of the prototype software. The conclusion and evaluation must be supported with literature review, citations and references.

The pass mark for the course work is 40% and students are required to gain a pass.

2.0 Course Work Requirement:

Each student or the group of **3 to 4 students** must select one case study in section 8 and inform the Module tutor. Thereafter the group of students must explain the business operation in the case study and perform the tasks listed below: Sections 8 are the case studies to be used as a guide or modified for the course work.

Tasks

Select any one of the case studies in Section 8 and briefly explain the core business operations or processes in it and carry out the following tasks.

1. Select and rewrite or modify the chosen case study

Select one of the case studies in section 8. You can update or modify the selected case study to correspond with the business processes in an enterprise.

2. Model the Business Processes (Group)

Use the updated or modified case study, to develop business process model BPMN **using Bonita software package**. The BPMN model, must clearly identify all the processes in the case study and model them using the appropriate business process notations. Each Group must review the BPMN lanes and develop a suitable BPMN by adding more activities.

3. Create Class diagram and Entity Data Model ERD (Group)

Develop a class diagram for the case study and convert the class model into entity data model (ERD) using the correct UML to ERD principles. Identify the main interface classes and entity data classes to be use for the practical implementation of the portal (i.e. on Bonita, Cloud or Website based). Use StarUML or Yworks CASE Tool for the design.

4. Design Enterprise Cloud Architecture model for your Case Study (Group)

Draw an Enterprise Cloud Architecture (ECA) model for the selected case study. The proposed (ECA) model must be based on the selected updated case study. The group members must use the Bonita BPMN, class and entity data model above as a guide to produce Cloud Architecture (ECA). The (ECA) model must comprise of the On premise system, AZURE or AWS systems with VPC and connection between On premise and off premise Systems. The proposed ECA model must address the connectivity problems in your selected case study and use the Draw IO, Visio or any CASE Tool. The Cloud Architecture model must be based on the selected Case Study.

5. Implementation of Portal CSS prototype software (Group)

Implement one prototype system 1. Using Either Website (SOA) or 2. Cloud (COA) based Client Server System (CSS) for the selected Case study. The CSS must comprise of Interface(s), with either SOA website or Cloud Systems (i.e. Azure or AWS), which are connected to the database backend using the models and results from Task 2, 3 and 4. The

implemented CSS, must perform and execute the main business processes and services specified in the Tasks 2 and 3.

6. Agile Project Management (Group)

Use Agile ScrumDesk software package to manage the group performance. Each group member's role and activities must be shown ScrumDesk dashboard and shared with the Tutorial Lecturers.

7. Individual Reflective Evaluation

Write an evaluation for the design and implementation of the prototype software. In your evaluation, analyse the work done on AWS and AZURE cloud technologies used for the design and implementation of the CSS. Develop a test plan and use it as guide to conduct the appropriate test for the verification of the information system. Support your write-up using relevant references and citations.

8. Group Conclusion

Write a group conclusion, which summaries the work done

Useful Tips:

Each group of a max of 4 students should demonstrate the appropriate use of notations in the business process model, class diagram, ERD and Enterprise Cloud Architecture model. The business process and Cloud Enterprise Architecture models must have titles and key of the elements used. The practical implementation section must have screen shots with figure numbers and explanations.

Class modelling

Use the updated case study to develop a class and data entity diagrams for the information system. This must include

1. The relevant classes and entities in the problem domain and the attributes from the case study.
2. All possible class (objects) and their names in the model;
3. Indicate the full attributes and operations with parameters for each class;
4. All static associations between classes, and their cardinalities.

Data Entity model or ERD

1. The data entity diagram must include all the entities in the problem domain and their corresponding ERD.
2. The ERD must show how the classes were converted into entities for the ERD with detail explanation.

Develop a simple working C/S prototype for the system

1. Implement a Website or Cloud Systems with Interface based on your class diagram using website software package and **MySQL** or **MS SQL Server** or any website development package of your choose. The backend must match the data entity diagram (ERD) on a database platform.
2. Connect the Web pages to the database, using suitable middleware.
3. Test the prototype Client Server software with two Interfaces and ensure that all the main functionalities in the business process model and methods in the Interface class (GUI) are operational.

Individual Reflective Evaluation and Group Conclusion

Evaluation of the cloud portal technology and techniques used:
Each student must conduct a literature review on the cloud portal technology used for the design and development task.

1. Write Individual reflective evaluation on the techniques used in the course work and justify why the techniques were chosen.
2. Analyse the work done using the AWS and AZURE cloud technologies. Support your evaluation with practical lab exercises.
3. The work must have references with in-text citations using Harvard Referencing system.
4. The Group conclusion must summarise the work done.

3.0 Submission.

The written report for the course work and the practical work must be submitted online on or before

Deadline

Your work must be submitted on **1st /May/2026**, using the submission link available on Moodle. In addition, an electronic copy of your work must be uploaded to the TURNITIN on Moodle on or before the submission date else the assignment will have failed.

4.0 Documentation guidelines and other requirements;

The Report must include the following

1. Title
2. Introduction
3. Brief overview of the selected Case study of an enterprise.
4. Write an updated case study using the business processes in the Case study and collect data that will be used for the course work.
5. Modelling and implementation Tasks
    - Business process model
    - Class diagram
    - Data entity model.

- Enterprise Cloud Architecture model

6. Working prototype software of the selected Case study of an enterprise.
7. Indication of Test results carried out (Test Plan with results)
8. Project Management – using Agile Scrum package
9. Evaluation must reflect the work done supported by literature review.
10. Conclusion
11. Appendices (include screen shots, test results etc.)
12. References

NOTES

You must create all diagrams using appropriate Case Tool and develop the Enterprise Information System Software with the appropriate IDE. You should include all the assumptions you have made while undertaking the tasks together. Your work must have an assignment header sheet and make sure that your student ID numbers are clearly legible.

5.0 Assessment Criteria

1. A high level of understanding of the subject must be demonstrated;
2. Demonstrate a wide and thorough understanding of the Enterprise Architecture modelling and software development techniques.
3. a high level of analytical ability;
4. imaginative design and application of suitable solution(s)
5. effective use of C.A.S.E. tool and web development software IDE;
6. An ability to analyse concepts and apply them in practice.
7. Your evaluation must indicate the analysis of the cloud-based portal technology and techniques used for the design and implementation of the CSS.

6.0 ASSESSMENT FEEDBACK

1. Feedback on your assessment will be available in four working weeks from the submission date.
2. Please refer to section 9 for specific details of the assessment.
3. The Feedback details on Moodle will include: -

- Comments on the models and diagrams.
- Comments on the relationship between the updated case study and models.
- The business processes and classes in the updated case study match the Enterprise Cloud Architecture model.
- Development and implementation of either Web or Cloud application software.
- Discussion and justification of the selection of the web or cloud portal technology and how it was used for design and implementation of CSS.
- Individual Reflective Evaluation and Group Conclusion.
- The use of in text citations and Harvard references.

7.0 Rules:

1. The report must not exceed **2500 +/- 10%** words; diagrams will not be included in the word count.
2. All materials must be original in the sense that you prepared them.
3. At least 25% of the work must demonstrate your views in the form of discussion, evaluation and conclusions.
4. All sources must be declared. Failure to state the source of any part will be penalised according to the University's regulations.
5. There should be a full reference list at the end of the report.

8. Case Studies to be used as guide:

The case studies below can be modified or updated and used as the case study for the course work.

Case Study1: Kings Estate Agent - Property Management Information System. The Kings Estate agent is an enterprise with many branches in UK. The enterprise wishes to create an information system to monitor each of its assets such as Servers, PCs, printers, cars, scanner, houses, flats, Apartment, desk etc. Assets are divided into a number of categories, such as buildings, computers and furniture. An asset is assigned to a staff. On a regular basis, staff in the Accounts Department checks each asset to determine its current market value and records the date and the current value of the asset. As a result of the assessment, the enterprise may decide to sell the asset. Also, on a regular basis, maintenance is carried out on each asset. In some cases, the maintenance is carried out by the staff, but in others, the maintenance of the asset is outsourced to an external maintenance company. Each staff has a specific role, which defines their charging rate and each staff can work on several assessment associated with the same maintenance.

The agency wants to create a web-based computer system that allows employees and external companies to access the company's portal (i.e. using Web site SOA, Azure or AWS cloud-based system). Model, design and implement client web pages that can access the database using MS SQL Server or any other web or cloud development package. The database must be designed from the class model and the entity data model using MySQL MS SQL Server, Oracle, any database and any AI or ML components (**not MS Access**).

Case Study 2. Consultancy Management System

A consultancy organisation wishes to create an information system to help manage its projects. Each project is for a specific customer and has a nominated project engineer who is supervised by project manager. The project is divided into a number of work packages and staff cost, staff time and expenses against a work package. Each staff has a specific role, which defines the charging rate for the customer. During the project duration, a staff can work on several work packages associated with the same project. In addition, most, but not all, work packages have a number of associated documents as deliverables, each of which may be written by more the one staff. Also on most

the projects are carried out in house by the engineer but in others, the project is outsourced to an external company.

The organisation wants to create a web-based computer system that allows employees and external companies to access the company's portal (i.e. using Web site SOA, Azure or AWS cloud-based system). Model, design and implement client web pages that can access the database using MS SQL Server or any other web or cloud development package. The database must be designed from the class model and the entity data model using MySQL MS SQL Server, Oracle, any database and any AI or ML components (**not MS Access**).

Case Study 3. Programme Management System

A training company wishes to create an information system of its Programme information. The company delivers a number of seminars and training courses. Each course is delivered by two members of staff at some location (such as internal seminar room or Greenwich Hotel Suite). The fees vary for each course and on the number of delegates a company sends. For example, if a company sends one person, the charge may be £5000. If the company sends two people, the first may be charged £5000, but the second may be charged £2500. The course can be attended by a number of delegates, subject to some upper limit for the course. A delegate can register as an individual or through his or her company. The name of the employee who registers the delegate is recorded. An invoice is sent either to the delegate or to his or her company. Most courses are carried out in house by the staff but in others, the courses are outsourced to external companies.

The company wants to create a web-based computer system that allows employees and external companies to access the company's portal (i.e. using Web site SOA, Azure or AWS cloud-based system). Model, design and implement client web pages that can access the database using MS SQL Server or any other web or cloud development package. The database must be designed from the class model and the entity data model using MySQL MS SQL Server, Oracle, any database and any AI or ML components (**not MS Access**).

Case Study 4: -Patients' Record for a CVD Clinic

A group of cardiovascular specialists run a CVD clinic in West London. The patients' information is currently collected on forms that are to be filled by the patients and examined by the staff before the patient reach to the doctor. There are many different forms for different reasons such as hypertension or hypotension consultation after specialist examination of the BMI, Blood Pressure and diabetics etc. The current system was examined and all aspects of the business activities, and the way they functioned were observed. The manager, every category of staff and some patients were interviewed. Sample forms that are given to patients were collected and the information asked for, and its purpose, were observed. Most of the forms, had 2 sections, the first section filled in by the patient before cardiovascular (CVD) examination and the second section was completed by the Consultant after examination and appointment session. There was a lot of information requested from registering patients, booking appointment etc. This was a common source of error that

increased waiting times. An information system can tackle the problem by automating the filling of certain fields; adding in drop-down menus and setting defaults. Finally, the Consultant gives the patient the requisite Prescriptions.

The CVD Clinic wants to create a web-based computer system that allows employees and external companies to access the company's portal (i.e. using Web site SOA, Azure or AWS cloud-based system). Model, design and implement client web pages that can access the database using MS SQL Server or any other web or cloud development package. The database must be designed from the class model and the entity data model using MySQL MS SQL Server, Oracle database and any AI or ML components (**not MS Access**).

Case Study 5: - State University

State University is building a Web-based registration system where 90% of all registration takes place during a 1-week period each semester. The State University has 35,000 students who must register. As a Web-based system, registration can take place 24/7 (24 hours a day, 7 days a week). However, a more reasonable profile is shown below (usage as number of registrations and hours on 24-hour clock). Each registration is approximately 400 kb. The university has decided to create an on-line course registration Centre. The on-line system allows prospective students to logon to the web based online system, select a course and register on it. There courses are made up of MSc IT and BIS. Each course has Course ID, name and description with "get Course ID" and "edit description" methods. To register on a course the prospective student must fill in their personal details on the online application form. The admission lecturer checks the application forms for the qualification then the admission officer verifies the applicants' personal details. After this procedure is completed the student is sent a username and password to complete the registration process. Users can also visit the on-line course registration site to browse the course and collect information.

The University wants to create a web-based computer system that allows employees and external companies to access the company's portal (i.e. using Web site SOA, Azure or AWS cloud-based system). Model, design and implement client web pages that can access the database using MS SQL Server or any other web or cloud development package. The database must be designed from the class model and the entity data model using MySQL MS SQL Server, Oracle, any database and AI or ML components (**not MS Access**).

Case Study 6: - State Clinic Health Information System

Suppose you are working on a project to develop a new electronic medical records system. The clinic has three doctors, ten nursing staff, and twenty administrative and clerical staff. The clinic has seven examination rooms and a laboratory. You are tasked with estimating the hardware costs associated with the new system. The system is a client-server system, with an Oracle database. Each examination room needs a computer, the laboratory, and approximately five computers for the administrative staff. The clinic has no existing network infrastructure, so that will need to be estimated as well. The

Clinical Centre has 5 treatment rooms (i.e. numbers 1-5), 6 -15 are open ward bedrooms and 16 - 45 are private bedrooms. When patients visit the clinic, the clinic manager books them into any available room they request. The clinic manager records the patients' names and personal details along with the payee's credit card details. The payee's details are entered as PRIVATE if the patient is the payee or the name of the company. The cost of visiting a doctor and receiving treatment in the treatment room is £100. Each treatment room is managed by a doctor and two staff nurses. The cost of a private bedroom is £50 and £40 for the open ward bedroom per week. Finally there is only one set of mobile operating equipment in the clinic that must be moved between treatment rooms.

The Clinic wants to create a web-based computer system that allows employees and external companies to access the company's portal (i.e. using Web site SOA, Azure or AWS cloud-based system). Model, design and implement client web pages that can access the database using MS SQL Server or any other web or cloud development package. The database must be designed from the class model and the entity data model using MySQL MS SQL Server, Oracle database and any AI or ML components (**not MS Access**).

Scenario 7: - University of London

The University of London is a school in London. As like other schools, there is an attendance policy the students have to follow. The student's absence is currently collected by making a student fill out a form. The absence form is then examined or reviewed by the school, where no feedback to the student is received, unless the student sends a mail. The reason is that the student has to use a third-party program in order to fill out the form. As a result of that, there are no ways for the student to know if an absence form was accepted or rejected by the school. The school wants to create a feature that allows students to register their reason for absence and receive feedback, by submitting a form connected to their student ID that exists in Univesity of London's database.

The process starts when the student first completes a form and submit it. The submitted form will then be received by the teacher, which will either approve or reject it. If the teacher decides to approve the form, two messages will be sent, one that goes back to the student, and another that is sent to the school. The first message that goes to the student includes feedback and will also confirm the acceptance for the student. The second message is for the school to update the student's record. However, if the form was rejected, the school would be notified by the teacher, and it will check the record of the student whether the student has a lot absence or not. In case of major absence, there will be a warning letter sent to student. If there is close to no absence, a mail of why the form was rejected will be sent to the student.

The University wants to create a web-based computer system that allows employees and external companies to access the company's portal (i.e. using Web site SOA, Azure or AWS cloud-based system). Model, design and implement client web pages that can access the database using MS SQL Server or any other web or cloud development package. The database must be designed from the class model and the entity data model using MySQL

MS SQL Server, Oracle, any database and any AI or ML components (**not MS Access**).

Case Study 8: Hardware Request Management System.
This University computing hardware department request system has different branches in different campus in London. The enterprise wishes to create an information system to monitor each of its assets such as Servers, PCs, printers, scanner, desk etc. Assets are divided into a number of categories, such as computers and Software. The hardware asset request process is assigned to a staff. On a regular basis, staff in the computing Department checks each asset to determine its current market value and records the date and the current value of the asset. As a result of the assessment, the University may decide to sell the asset. Also, on a regular basis, maintenance is carried out on each asset. In some cases, the maintenance is carried out by the staff, but in others, the maintenance of the asset is outsourced to an external maintenance company. Each staff has a specific role, which defines their charging rate and each staff can work on several assessment associated with the same maintenance.

The University wants to create a web-based computer system that allows employees and external companies to access the company's portal (i.e. using Web site SOA, Azure or AWS cloud-based system). Model, design and implement client web pages that can access the database using MS SQL Server or any other web or cloud development package. The database must be designed from the class model and the entity data model using MySQL MS SQL Server, Oracle database and any AI or ML components (**not MS Access**).

Other topics include the following:

Case Study 9: UEL e-Library Systems

Case Study 10: Diabetic Health Care Systems

Case Study 11: UEL Electronic Cafeteria Systems

Case Study 12: Railway Electronic Booking Systems

Section 9: Semester A
2025/2026 CN6001/CN5023: Marking Criteria for Group Work

- The actual group work documentation and practical implementation is 30%;

Assessment Criteria	Excellent	V Good	Satisfactory	Fail	Mark/Comment
	70%-100%	50%-69%	40%-49%	0%-39%	
Introduction: Overview of the selected case study. Discussion on the operations and main business processes. **(5 Marks)**					
Business Process Model Create a detailed business process model using Bonita BPMN with explanation. **Group (10 marks)** **Class diagram and ERD** Develop a class diagram with Software and transform it into entity data model (ERD) with explanation. **Group (10 marks)**					
Enterprise Cloud Architecture model Draw Enterprise or Cloud Architecture models for the case study with explanation. **Group (10 marks)** **CSS Software** Develop prototype software using desktop and website or Cloud with backend driven by database. **Group (30 marks)** **Agile Project management** Use of Agile ScrumDesk to manage the Group project. **Group (7 marks)**					

Individual Reflection: An evaluation of work done on AWS/AZURE cloud technologies and Testing. Supported with in-text citations and Harvard referencing. **(10 marks)**					
Group Conclusion: Must be relevant to the content and critical. Highlight all major points and issues.**(10 marks)**					
Presentation: Well organised and structured report with sections. **(4 Marks)** **References:** Must have end of C/W references with correct Harvard Referencing system **(4 Marks)**					

First Marker_____% Date: _____

Second Marker _____% Date: _____

FINAL MARK _____% Date